

WALMART SALES PERFORMANCE DASHBOARD

Using Microsoft Power BI

Project Report

Submitted By

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Tool Used : Microsoft Power BI Desktop

Introduction

For this project, I developed an interactive Sales Performance Dashboard in Power BI using Walmart's weekly sales dataset. I selected this dataset because it contains useful information such as store-wise sales, dates, fuel prices, and holiday status, which provided an opportunity to analyse sales performance from different perspectives.

The main purpose of the dashboard was to convert the raw Walmart sales data into a simple and interactive visual report that can help users understand sales performance quickly. Instead of looking through a large dataset, the dashboard allows users to view important sales information through KPI cards, charts, and interactive filters.

The dashboard provides an overview of total sales, average weekly sales, transaction count, sales trends, store-wise performance, fuel price comparison, and holiday versus non-holiday sales.

Objectives

The main objectives of developing this dashboard were:

- To analyse Walmart's weekly sales performance across different stores.
- To create an interactive and visually appealing dashboard using Power BI.
- To identify stores with higher and lower sales performance.
- To analyse how weekly sales changed over the given period.
- To compare sales during holiday and non-holiday periods.
- To examine the relationship between fuel prices and weekly sales.
- To provide important sales information through KPI cards for quick understanding.

Dataset Description

The dataset used for this project is the Walmart Sales Dataset, which contains weekly sales information from Walmart stores.

The major fields used in the analysis were:

- **Store** – identifies the Walmart store.
- **Date** – represents the week in which the sales were recorded.
- **Weekly Sales** – represents the total sales recorded for a store during a particular week.
- **Fuel Price** – represents the fuel price associated with the particular period.
- **CPI** – represents the Consumer Price Index.
- **Unemployment** – represents the unemployment rate.
- **Holiday Flag** – indicates whether the particular week was a holiday week or a non-holiday week.

These fields provided sufficient information to analyse sales performance across stores and over time.

Data Preparation

Before creating the dashboard, I first prepared the dataset in Power BI Desktop.

The main steps I followed were:

- Imported the Walmart sales Excel file into Power BI.
- Checked the columns and their data types.
- Changed the Date column to the correct date format.
- Checked that Weekly Sales and Fuel Price were recognised as numerical values.
- Checked the Store and Holiday Flag columns for filtering and analysis.
- Created the required DAX measures for the KPI cards.

After preparing the data, I started creating the different visuals for the dashboard.

DAX Measures Used

I created a few simple DAX measures to display important information on the dashboard.

- **Total Weekly Sales**

This measure calculates the total sales from all the available records.

- **Average Weekly Sales**

This measure calculates the average sales for the given data.

- **Transaction Count**

This measure counts the number of sales records available in the dataset.

These measures are displayed as KPI cards at the top of the dashboard. They make it easy to understand the overall performance without going through the complete dataset.

Dashboard Components

I used different types of charts in the dashboard so that each chart could show a different part of the sales data.

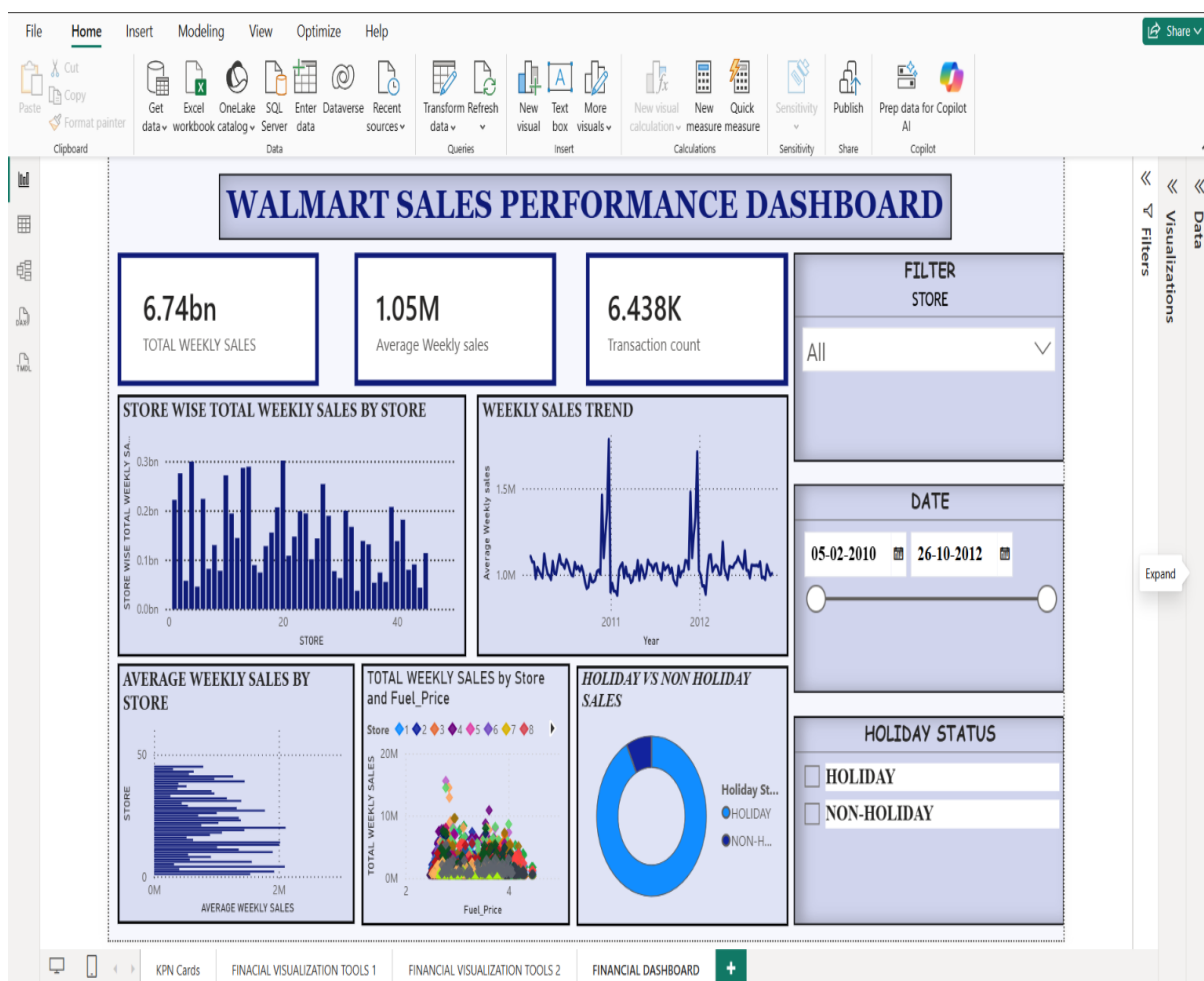
1. KPI Cards

The first section of the dashboard contains three KPI cards.

They show:

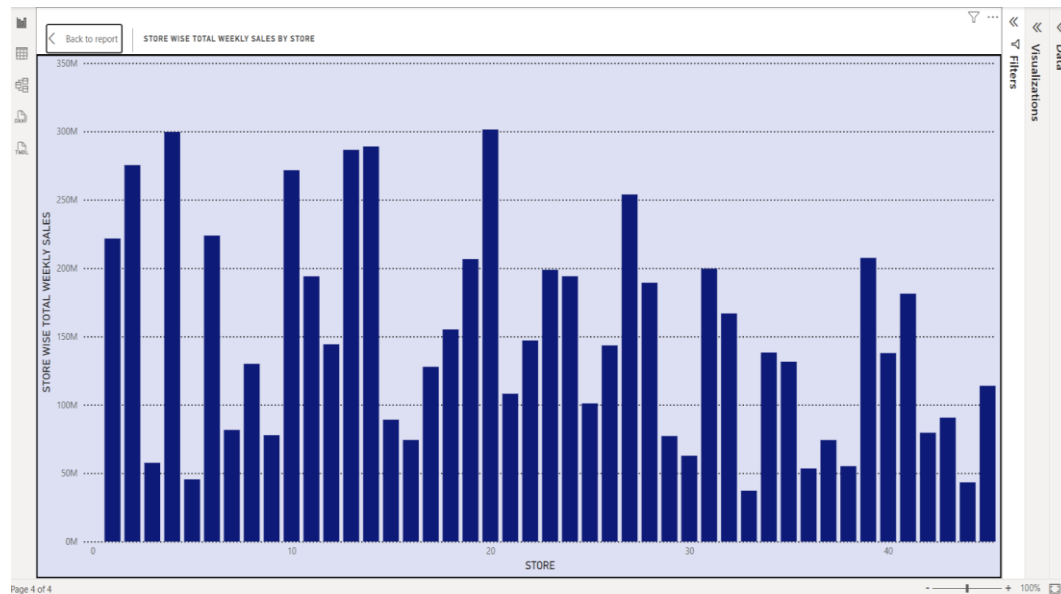
- Total Weekly Sales – 6.74bn
- Average Weekly Sales – 1.05M
- Transaction Count – 6.438K

These cards give a quick summary of the Walmart sales data.



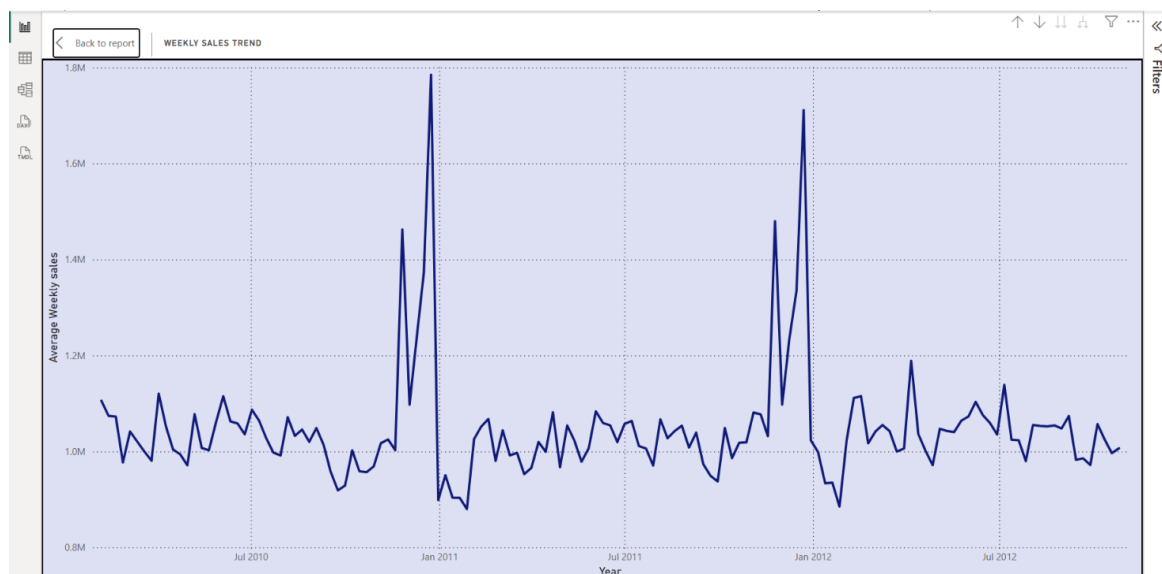
2. Store-wise Total Weekly Sales

I used a column chart to show total weekly sales for different stores. This chart helps compare the sales performance of the stores. By looking at the chart, we can easily see that some stores have higher sales while some stores have lower sales. This makes it easier to identify the better-performing stores.



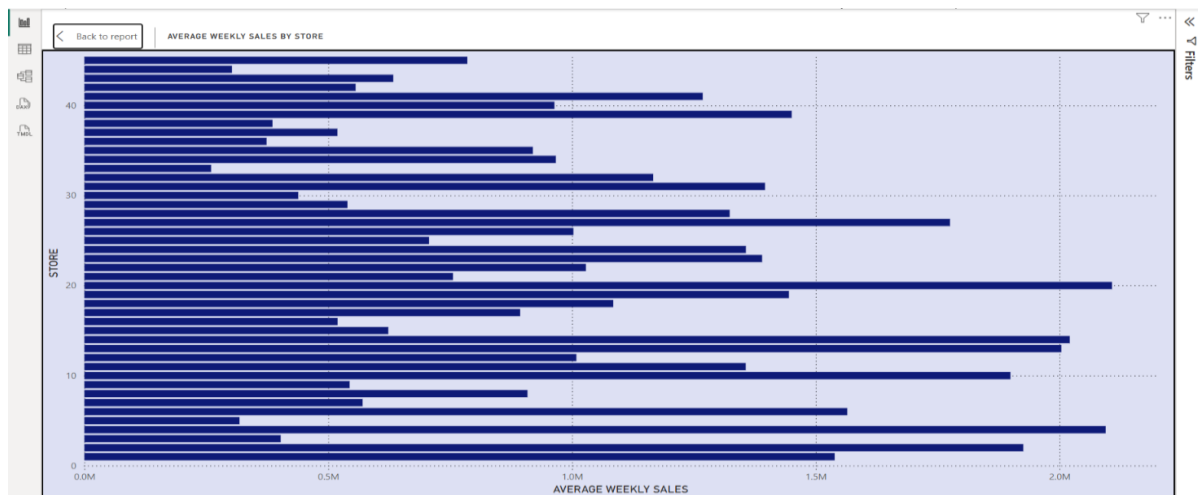
3. Weekly Sales Trend

I used a line chart to show the weekly sales trend over time. This chart helps us understand how Walmart's sales changed from one period to another. The sales are not the same every week. There are some periods where sales increase significantly and some periods where sales decrease. The chart also helps us notice higher sales during certain seasonal periods.



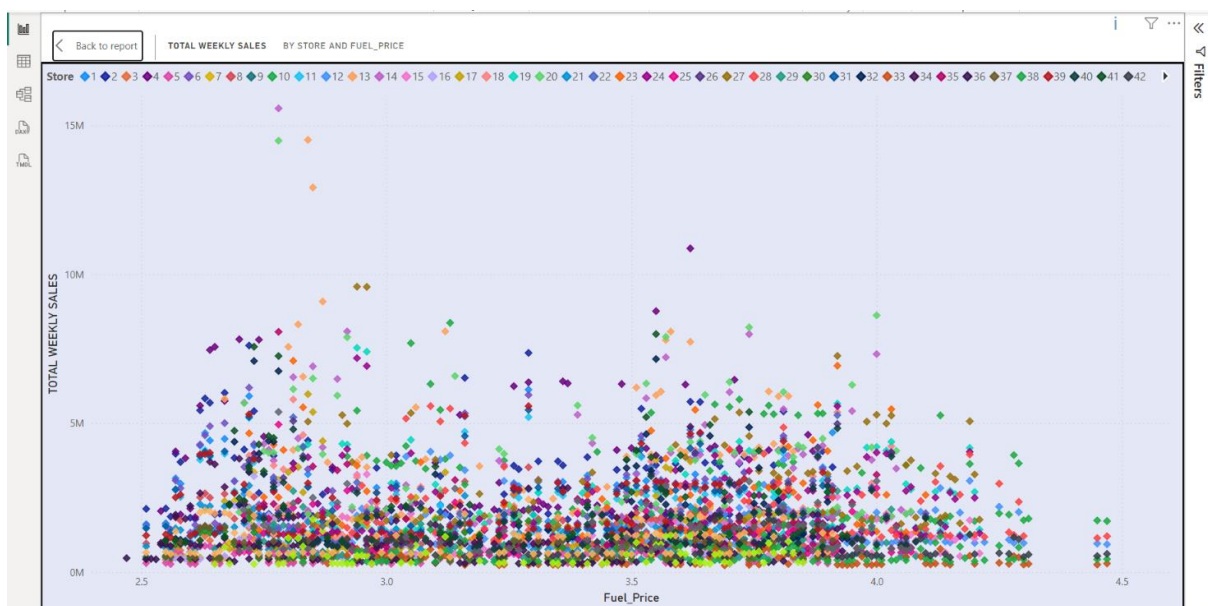
4. Average Weekly Sales by Store

A horizontal bar chart is used to show the **average weekly sales for each store**. This helps us compare the average performance of the different stores. Instead of looking at total sales, this chart shows the average sales made by each store, making it easier to understand the regular sales performance of individual stores.



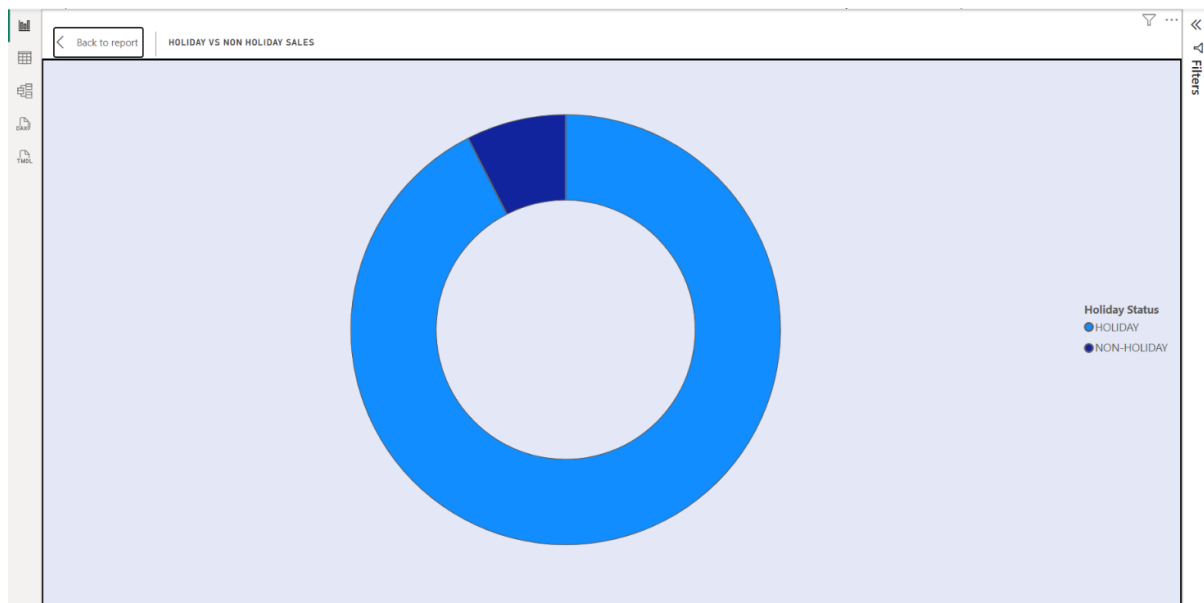
5. Total Weekly Sales by Store and Fuel Price

I used a scatter chart to compare total weekly sales with fuel prices for different stores. The chart contains many scattered points and does not show a clear pattern between fuel price and weekly sales. From this visual, we can understand that fuel price does not appear to have a strong direct relationship with Walmart's weekly sales in this dataset.



6. Holiday vs Non-Holiday Sales

I used a doughnut chart to compare sales during holiday and non-holiday periods. The chart shows that non-holiday sales make up a larger portion of the total sales. This does not necessarily mean that holiday weeks have lower sales. There are simply many more non-holiday weeks in the dataset, so their total contribution is larger.



Slicers Used

To make my dashboard interactive, I added different filters or slicers.

- **Store Filter**

The Store filter allows the user to select a particular store or view all stores.

- **Date Filter**

The Date filter allows the user to select a particular date range and analyse sales during that period.

- **Holiday Status Filter**

This filter allows the user to select Holiday or Non-Holiday sales.

The main advantage of these slicers is that the charts and KPI values change automatically when a filter is selected.

This makes the dashboard more useful because the user can explore the data according to their requirement.

The image shows a vertical filter panel with a light blue background and a thin border. It contains three distinct filter sections:

- FILTER STORE:** A dropdown menu with a white background and a blue border. The text "All" is visible on the left, and a blue downward-pointing chevron is on the right.
- DATE:** A date range selector. It features two white input boxes with blue borders. The first box contains "05-02-2010" and the second contains "26-10-2012". Each box has a small blue calendar icon to its right. Below these boxes is a horizontal line with two white circular handles, indicating a range slider.
- HOLIDAY STATUS:** A section with two radio button options. Each option consists of a small white square with a blue border followed by the text "HOLIDAY" and "NON-HOLIDAY" respectively.

Key Findings

After analysing the dashboard, I found the following:

- ✓ The total weekly sales are approximately 6.74 billion.
- ✓ The average weekly sales are approximately 1.05 million.
- ✓ The dashboard contains around 6.438K sales records.
- ✓ Sales are different across the various Walmart stores.
- ✓ Some stores have much higher total and average sales compared to other stores.
- ✓ Weekly sales change over time and do not remain constant.
- ✓ Non-holiday sales contribute a larger portion of the total sales.
- ✓ The scatter chart does not show a strong relationship between fuel price and weekly sales.
- ✓ Using the filters makes it easier to study the performance of a particular store or time period.

Conclusion

This project helped me understand how Power BI can be used to analyse and present sales data in a simple way.

By using the Walmart sales dataset, I was able to create different charts to understand store-wise sales, weekly sales trends, average sales, holiday sales, and the relationship between fuel prices and sales.

The KPI cards helped me understand the overall performance quickly, while the charts provided more detailed information. The slicers made the dashboard interactive and allowed me to explore the data based on store, date, and holiday status.

Overall, this project improved my knowledge of Power BI, DAX, data visualisation, and data analysis. It also helped me understand how raw data can be converted into a useful dashboard that makes information easier to analyse and understand.

References

- Microsoft Power BI Desktop
- Walmart Sales Dataset
- Microsoft Power BI Documentation